Recombinant DNA Technology

Module 2

Expected Learning Outcomes

2.1. Define recombinant DNA technology and explain how it is used to clone genes and manipulate DNA.

2.2. Compare and contrast different types of cloning vectors; describe their practical features and applications.

2.3. Discuss how DNA libraries are created and screened to identify cloned genes of interest.

The Big Idea!

Biotechnology, though often associated with modern laboratory breakthroughs, has roots that trace back centuries. Early forms of biotechnology were already practiced in agriculture and food production, but the field took a transformative leap with the discovery and manipulation of DNA. Since the 1970s, the emergence of molecular tools has revolutionized science, opening doors to understanding life at the genetic level. These advancements have not only deepened scientific knowledge but have also impacted medicine, agriculture, and industry. The evolution of biotechnology continues to shape the modern world, driving innovation and discovery across multiple disciplines.

Deepen Your Understanding!

This topic delves into the powerful techniques that define modern biotechnology, particularly recombinant DNA technology and genomics. It explores how scientists manipulate genetic material through DNA cloning to study gene structure, function, and expression. The discussion also covers various molecular tools and laboratory methods that enable gene isolation, modification, and transfer. Beyond gene-level studies, the topic introduces genomics and bioinformatics, which allow the analysis of entire genomes and their biological significance. Together, these areas highlight how molecular biotechnology continues to transform research, medicine, and our understanding of life itself.

RECOMBINANT DNA TECHNOLOGY AND DNA CLONING

The story of recombinant DNA technology begins with one of the greatest scientific discoveries in history—the **identification of the DNA double helix** by James Watson and Francis Crick. This breakthrough revealed how genetic information is stored and passed from one generation to the next. However, Watson and Crick could not have imagined how rapidly molecular biology would advance after their discovery. In the years that followed, many scientists devoted their efforts to understanding how DNA

functions as the genetic material of living organisms. By studying bacteria and viruses called **bacteriophages** (or phages for short), researchers uncovered key details about how DNA replicates and how certain enzymes control this process. One of these enzymes, **DNA ligase**, was found to join together small fragments of DNA, known as Okazaki fragments, during replication. This same enzyme would later play a vital role in connecting DNA pieces from different sources, a process that lies at the heart of recombinant DNA technology.

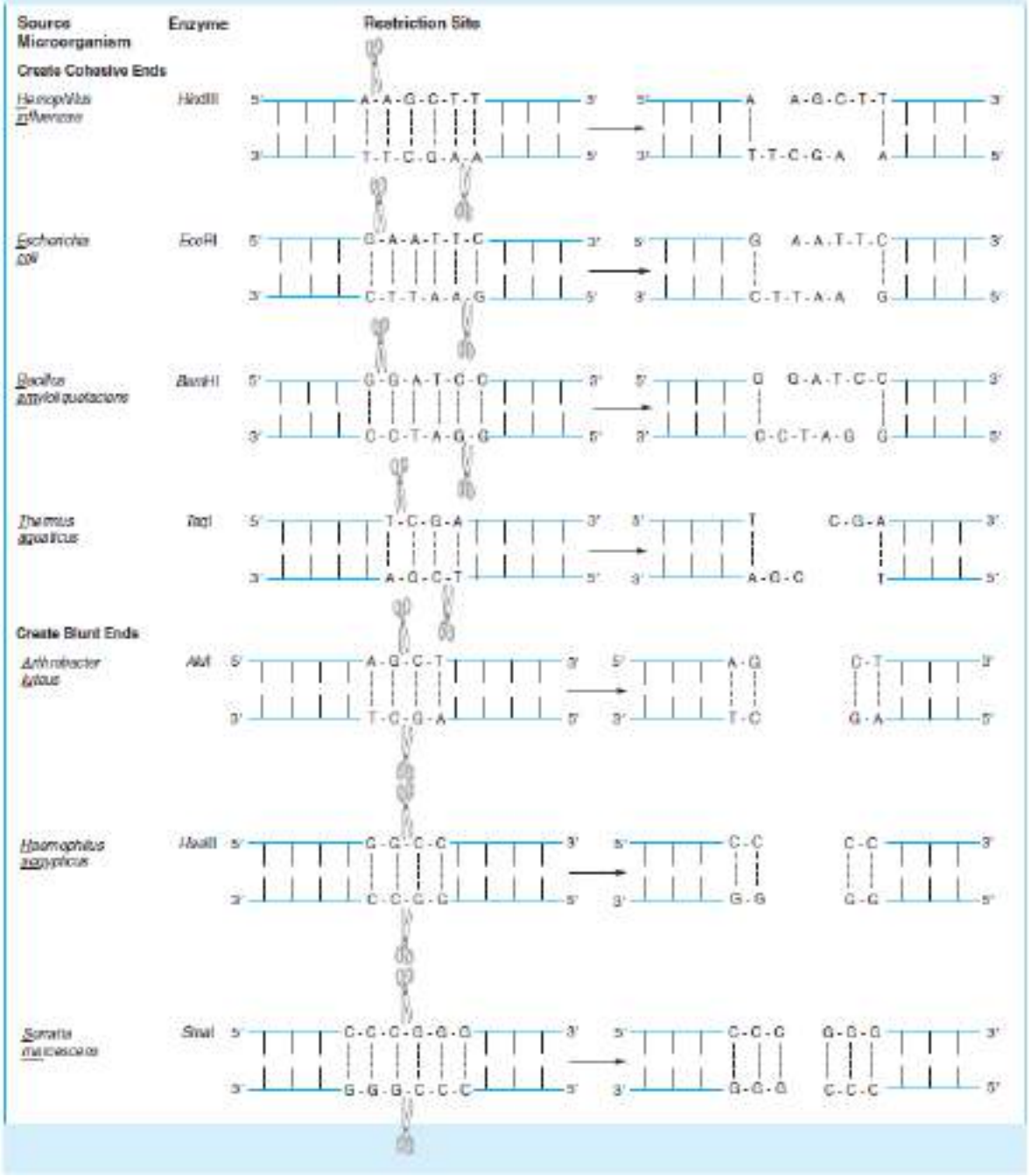
In the late 1960s, scientists began to imagine a new possibility: cutting DNA from one organism and joining it with DNA from another. This idea gave rise to **recombinant DNA technology**, which literally means combining DNA molecules from different sources to form new genetic combinations. This concept is closely related to **gene cloning**, which involves making identical copies of a particular gene, and **genetic engineering**, which focuses on altering the genetic makeup of organisms. Although these terms are often used interchangeably, they each emphasize a different aspect of genetic manipulation. The term clone comes from a Greek word meaning "twig," referring to a cutting used to grow an identical plant. In modern biology, a clone refers to a molecule, cell, or organism that is an exact copy of another. It is important to note that cloning a gene in the laboratory is very different from cloning an entire organism. Gene cloning focuses on copying small segments of DNA to study and use them for scientific or medical purposes.

RESTRICTION ENZYMES AND PLASMID DNA VECTORS

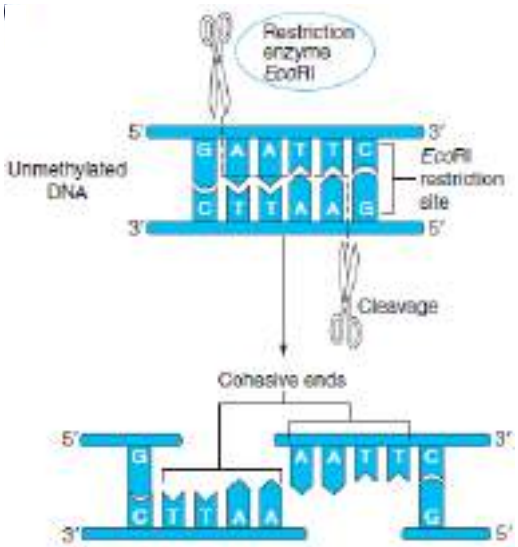
The dream of cloning genes became a reality in the early 1970s, when scientists made two critical discoveries: **restriction enzymes** and **plasmids**. Restriction enzymes are special proteins found in bacteria that protect them from viral infections. They work by cutting foreign DNA into smaller pieces, preventing viruses from taking over the bacterial cell. Because of this ability, Swiss scientist **Werner Arber** proposed that restriction enzymes act as a defense mechanism against invading bacteriophages. Later, **Hamilton Smith** at Johns Hopkins University isolated one of the first well-studied restriction enzymes, **HindIII**, from the bacterium *Haemophilus influenzae*. He demonstrated that this enzyme could cut DNA into specific fragments rather than randomly, showing that restriction enzymes recognize unique DNA

sequences called **restriction sites**. For his contribution, Smith shared the 1978 Nobel Prize in Medicine with Werner Arber and **Daniel Nathans**.

TABLE 1 COMMON RESTRICTION ENZYMES



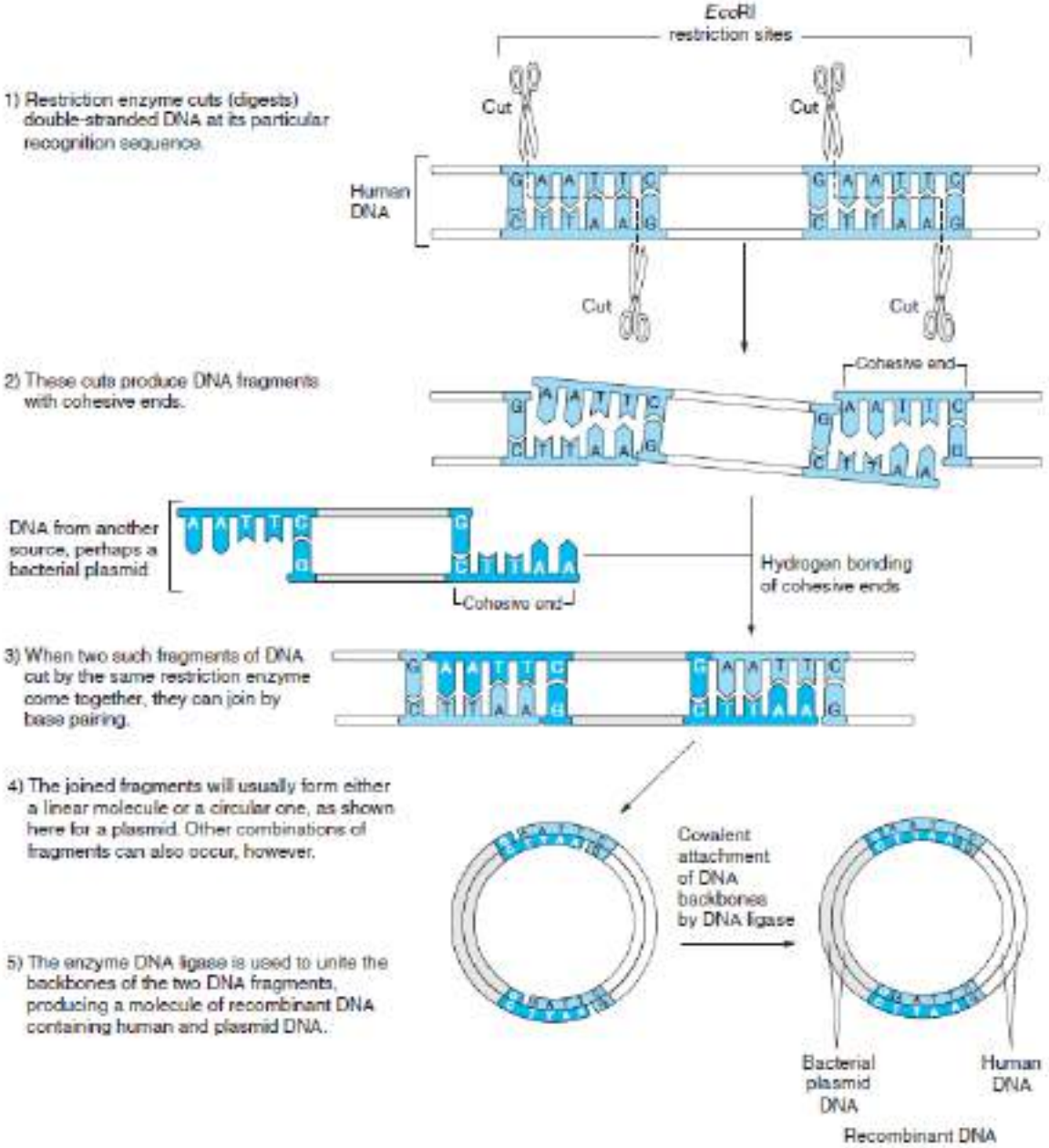
Restriction enzymes have since become essential tools in molecular biology. Each enzyme cuts DNA at a specific site, often a sequence of four, six, or eight base pairs. These sequences are usually **palindromes**, meaning they read the same backward and forward, like the word "madam." Some enzymes, such as **EcoRI**, produce fragments with overhanging single-stranded ends known as **sticky ends**, while others create **blunt ends** with no overhangs. Sticky ends are often preferred in gene cloning because they make it easier for scientists to join DNA fragments from different sources. When combined with **DNA ligase**, which seals the breaks in the DNA backbone, these enzymes allow researchers to "cut and paste" genetic material from any organism—bacteria, plants, animals, or even ancient remains—into a new host. In simple terms, restriction enzymes became the molecular "scissors" that made recombinant DNA technology possible.



PIONEERING EXPERIMENTS IN GENE CLONING

Using these new tools, scientists such as Paul Berg, Stanley Cohen, and Herbert Boyer revolutionized molecular biology. Paul Berg was the first to create recombinant DNA by combining DNA from E. coli bacteria with DNA from a primate virus called SV40. He used the enzyme EcoRI to cut both types of DNA and DNA ligase to join the fragments, forming a hybrid molecule. This experiment proved that DNA from different organisms could be joined together, earning Berg the 1980 Nobel Prize in Chemistry. Around the same time, Stanley Cohen and Herbert Boyer explored small circular DNA molecules called plasmids, which naturally occur in bacteria. Plasmids are separate from the bacterial chromosome and can replicate independently. Because of this property, Cohen suggested that plasmids could be used as vectors—carriers that can accept, copy, and transport foreign DNA.

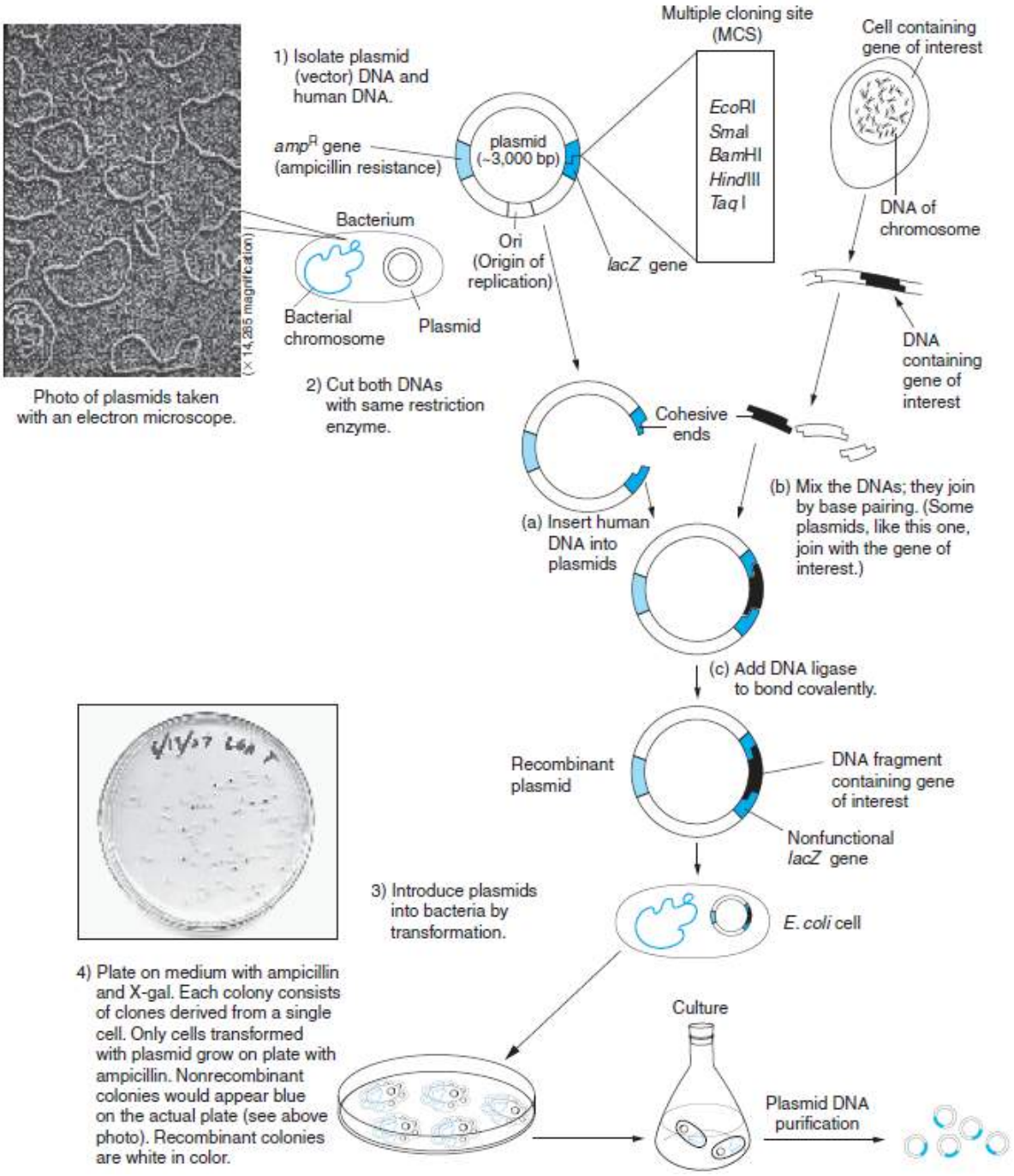
Cohen and Boyer's groundbreaking work in 1973 produced the first plasmid vector, called pSC101, which contained a gene for antibiotic resistance and specific restriction sites for DNA insertion. Using EcoRI and DNA ligase, they successfully joined DNA fragments from different plasmids, creating recombinant plasmids that could replicate inside bacteria. This experiment is considered the birth of modern biotechnology because it provided a reliable method to clone and manipulate genes. Later, Cohen and Boyer also cloned DNA from the frog Xenopus laevis into bacterial plasmids, proving that genes from one species could function in another.



TRANSFORMATION OF BACTERIAL CELLS AND SELECTION OF RECOMBINANT BACTERIA

After learning how recombinant DNA can be created in the laboratory, it is important to understand how this DNA can be introduced into living bacterial cells. This process, called **transformation**, was further developed by Stanley Cohen and his team to make gene cloning possible. Cohen discovered that bacterial cells could be made more receptive to foreign DNA if they were first treated with a calcium chloride solution, kept cold on ice, and then briefly exposed to heat. This heat shock allowed the plasmid DNA to slip through the bacterial cell wall and enter the cell. Once inside, the plasmid could **replicate independently** and express its genes, including those that make bacteria resistant to antibiotics. Later, scientists developed a more advanced technique called **electroporation**, which uses a quick burst of electricity to open small holes in the bacterial cell wall. This allows DNA to enter more easily and can even be applied to mammalian or plant cells. Despite these breakthroughs, transformation is still not perfectly efficient. Some plasmids close back on themselves without taking in foreign DNA, while many bacterial cells fail to take up any DNA at all.

To solve this problem, scientists designed clever ways to **identify which bacterial cells** have been successfully transformed with recombinant DNA. One of the earliest and most effective methods is called **antibiotic selection**. In this approach, bacteria are grown on agar plates that contain antibiotics such as **ampicillin**. Only bacteria that have taken in plasmids carrying an antibiotic resistance gene can survive and form colonies. However, antibiotic selection alone cannot distinguish between bacteria with a nonrecombinant plasmid (a plasmid that re-formed without foreign DNA) and those with a recombinant plasmid (a plasmid containing an inserted gene). To make this distinction, scientists use a method known as **blue-white selection**, which involves a gene called **lacZ**. This gene encodes an enzyme known as **β-galactosidase** (**β-gal**), which can break down lactose or similar compounds. When a gene is inserted into the lacZ site, the enzyme cannot be produced, and the reaction it normally triggers does not occur.

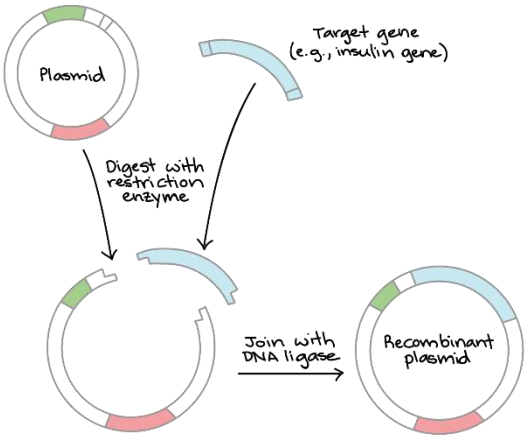


In blue-white selection, transformed bacteria are placed on agar plates containing both an antibiotic (such as ampicillin) and a special compound called **X-gal**. X-gal acts like lactose but turns blue when broken down by β-galactosidase. As a result, bacteria with nonrecombinant plasmids produce the enzyme and appear as **blue colonies**. In contrast, bacteria containing recombinant plasmids do not produce β-gal, cannot

break down X-gal, and therefore form **white colonies**. The white colonies are the desired ones because they contain recombinant DNA. This color-based system allows scientists to easily see which bacteria have taken in the new gene of interest. Each white colony represents **a clone** of genetically identical cells, all carrying the same recombinant plasmid.

Through transformation and blue-white selection, scientists can reliably identify and multiply bacterial cells carrying specific DNA inserts. This process is one of the most essential steps in **molecular cloning**, enabling researchers to produce large quantities of a gene for study or for practical use in medicine, agriculture, and biotechnology. These methods not only demonstrate the precision of genetic engineering but also highlight how simple visual cues—like the color of a bacterial colony—can reveal complex molecular processes at work inside living cells.

INTRODUCTION TO HUMAN GENE CLONING



The discovery of **restriction enzymes**, **DNA ligase**, and **plasmids** provided scientists with the essential tools to explore and manipulate genes from virtually any organism. When combined with transformation—the process of introducing foreign DNA into bacterial cells—these tools opened the door to a new era in biotechnology. Through **recombinant DNA technology**, scientists could now cut and join DNA fragments, insert them into plasmids, and allow bacteria to replicate these recombinant molecules. The inserted DNA, often called **insert DNA**, could be copied endlessly as bacteria multiplied, producing countless identical DNA molecules. This advancement allowed scientists not only to study genes but also to use bacteria as miniature factories capable of producing valuable proteins. When a cloned DNA fragment contained a gene that coded for a protein, the bacteria could be made to express that gene, generating the desired protein product in large amounts. This realization marked a turning point in molecular biology,

as it became clear that human genes could be cloned and expressed in bacteria for scientific and medical purposes.

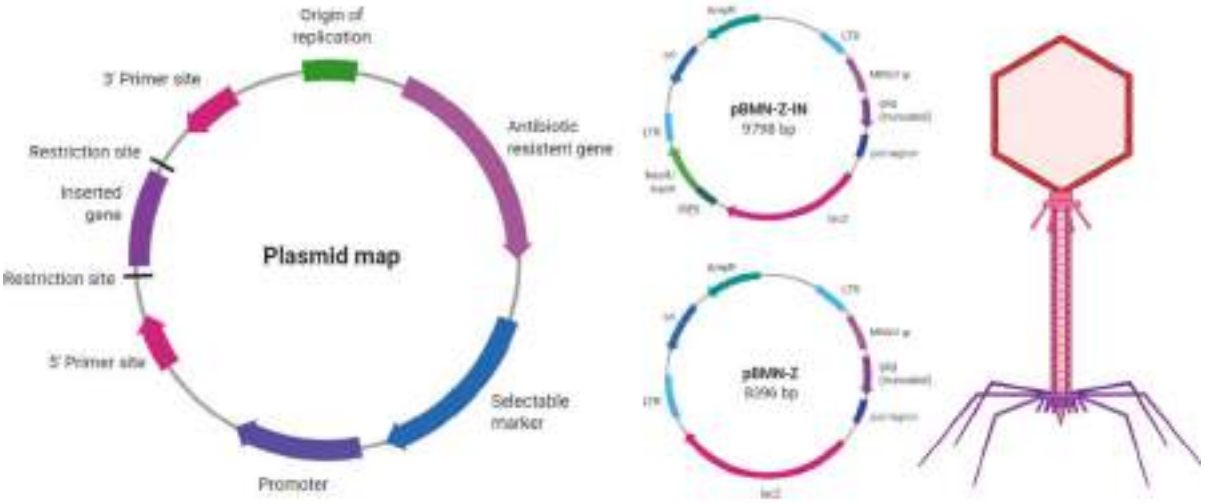
The success of this technology quickly led to some of the most important medical breakthroughs in history. One of the earliest and most celebrated achievements was the cloning of the **human insulin gene** in 1977 by scientists at **Genentech**, a biotechnology company founded by **Herbert Boyer** and **Robert Swanson**. Before this innovation, insulin for diabetic patients had to be extracted from the pancreases of pigs or cows—a process that was costly, limited, and sometimes caused allergic reactions. Using recombinant DNA technology, scientists inserted the human insulin gene into plasmids, transformed them into bacterial cells, and allowed these cells to produce human insulin naturally. This recombinant insulin, known as **Humulin**, became the **first genetically engineered product** approved by the **U.S. Food and Drug Administration** (FDA) in 1982. Its success was soon followed by the cloning of **human growth hormone**, which provided safe and effective treatment for children with growth disorders.

Before recombinant DNA methods were developed, isolating hormones like insulin or growth hormone required collecting tissues from animals or even human cadavers, which posed risks of contamination from viruses and other pathogens. Recombinant DNA technology solved these problems by providing a clean, efficient, and scalable way to produce human proteins in the laboratory. Today, hundreds of recombinant products are available, serving vital roles in **medicine**, **agriculture**, and **basic research**. Through human gene cloning, scientists have not only improved the quality of life for millions but have also deepened our understanding of how genes function and can be harnessed for good. With this foundation, the next section explores how various **DNA vectors** are chosen and designed to meet specific goals in genetic research and biotechnology.

WHAT MAKES A GOOD VECTOR?

As molecular biology continued to evolve, scientists began developing **better DNA vectors** to make gene cloning and expression more efficient. A **vector** acts as a carrier that transfers genetic material into a host

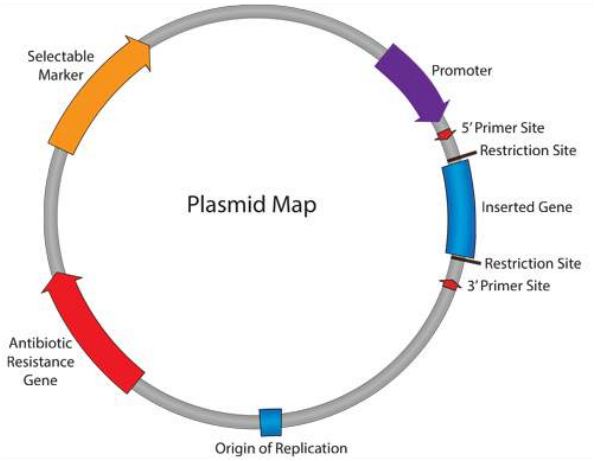
cell, allowing scientists to insert, replicate, and study genes. Since the creation of the first plasmid vector **pSC101** by Stanley Cohen, researchers have designed many improved versions with advanced features. Among all types, **plasmids** remain the most widely used cloning vectors because they are easy to manipulate, simple to introduce into bacterial cells, and convenient to isolate afterward. One of the earliest successful plasmid vectors, **pBR322**, contained genes for **ampicillin** and **tetracycline resistance** as well as several restriction sites that allowed for gene insertion. Over the years, new plasmids have been engineered to include additional features, making cloning faster, more reliable, and more versatile.



To be considered a **good vector**, a plasmid must have certain key characteristics that make it efficient and practical for use in the laboratory. These features ensure that DNA cloning and expression occur smoothly and accurately. The most desirable features of a DNA cloning vector include:

- **Small size** – The plasmid should be small enough to be easily separated from the chromosomal DNA of the host cell and simple to manipulate during experiments. Typically, it should be around **2,000** (**2kb**) to **20,000 bp** (**20kb**), but other type of vectors may contain up to **200kb** to **2,000kb**.
- **Origin of replication** (**ori**) – This is the site where DNA replication begins, allowing the plasmid to replicate independently of the host cell's chromosome. Some plasmids are high-copy-number plasmids, which can produce hundreds or thousands of copies per bacterial cell.

- **Multiple cloning site** (**MCS**) – The MCS is a short region containing recognition sites for several **restriction enzymes**. It allows scientists to insert DNA fragments without disrupting other important regions of the plasmid, providing flexibility in cloning genes from different sources.
- **Selectable marker genes** – These genes help identify bacterial cells that have successfully taken up the plasmid. Common examples include **ampicillin resistance** (**ampR**), **tetracycline resistance** (**tetR**), and the **lacZ gene**, which is used in **blue-white selection** to distinguish recombinant from nonrecombinant bacteria.
- **RNA polymerase promoter sequences** – These sequences allow RNA polymerase to transcribe cloned genes into RNA either inside the cell (**in vivo**) or in controlled laboratory reactions (**in vitro**). This feature is important for **protein synthesis** and studying gene expression.
- **DNA sequencing primer sites** – These are short DNA sequences that allow scientists to easily determine the **nucleotide sequence** of inserted DNA fragments.



These characteristics make plasmid vectors reliable and powerful tools for **recombinant DNA technology**. With these features, scientists can efficiently replicate, express, and analyze genes, paving the way for discoveries that continue to advance biotechnology, medicine, and genetic research.

TYPES OF VECTORS

In biotechnology, vectors are DNA molecules used as vehicles to transfer foreign genetic material into another cell, where it can be replicated and sometimes expressed. Just as a single screwdriver cannot handle every screw type, one kind of vector cannot serve every cloning purpose. Bacterial plasmid vectors were the first used in genetic engineering, but they have limitations. For example, plasmids can only carry small fragments of DNA, usually no larger than 6 to 7 kilobases (kb), and sometimes bacterial cells

cannot properly express proteins made from eukaryotic genes. To overcome these limitations, scientists developed several other types of vectors, each suited to different research applications.

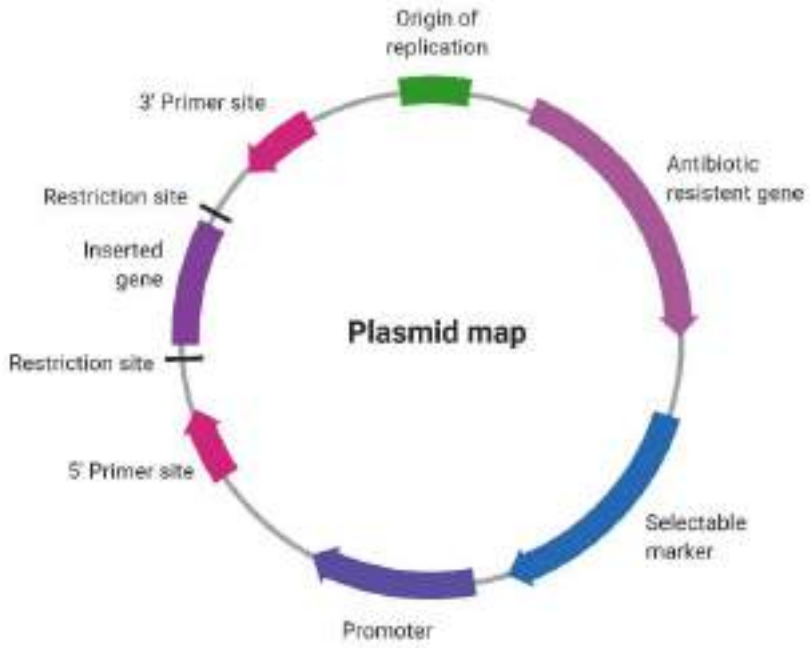
Below are the major types of vectors used in recombinant DNA technology:

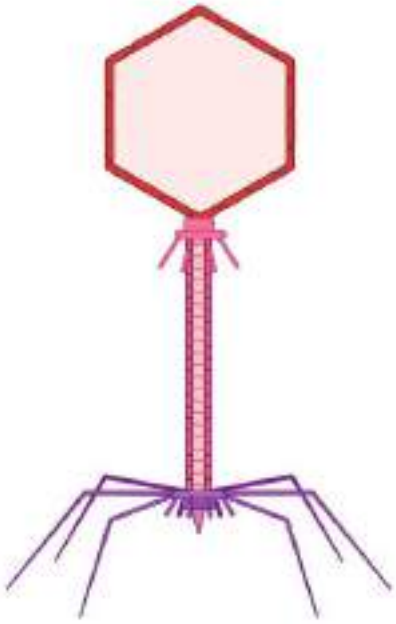
1. **Bacteriophage Vectors**

These are based on **bacteriophage lambda** (λ), a virus that infects *E. coli* bacteria. The λ chromosome is about **49 kb** long and can be engineered to carry foreign DNA fragments. When the recombinant λ infects E. coli, it forms **plaques**, or clear zones, on a bacterial lawn, representing areas where bacteria have been lysed (broken open) by the virus.

Key features:

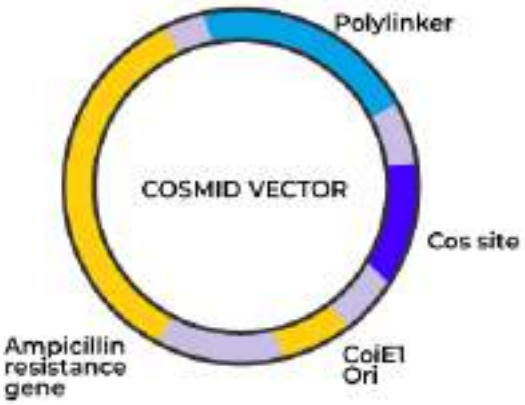
- Can clone DNA fragments up to **25 kb**, larger than plasmids.
- Use **cohesive** (**COS**) **sites** that help circularize the DNA during infection.
- Replicate through a **lytic cycle**, producing many viral particles.
- Not as commonly used today but important in early molecular biology research.





2. **Cosmid Vectors**

Cosmids combine features of plasmids and bacteriophages. They contain the **COS ends** of λ DNA, a plasmid **origin of replication**, and **antibiotic resistance genes**, but lack most viral genes. Once packaged into viral particles and introduced into *E. coli*, they behave like low-copy-number plasmids.

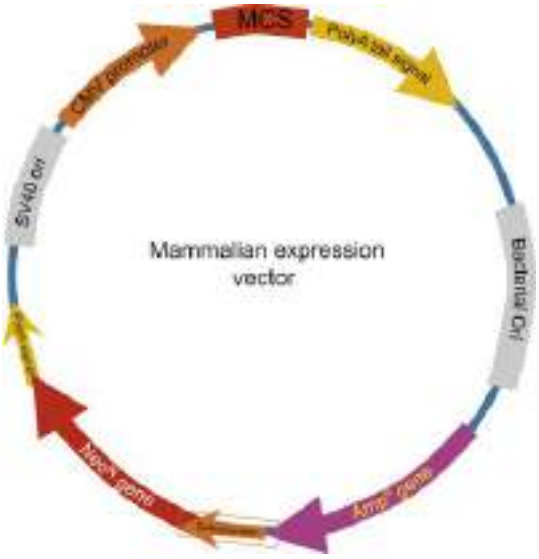


Key features:

- Can carry larger DNA fragments (around **20–45 kb**).
- Can be selected using **antibiotic resistance markers**.
- Now used less often due to newer, more efficient vectors.

3. **Expression Vectors**

These vectors are designed for the **production of proteins** inside host cells. They contain a **prokaryotic promoter sequence** next to the inserted DNA, allowing bacterial enzymes to produce large amounts of RNA and, consequently, the corresponding protein.



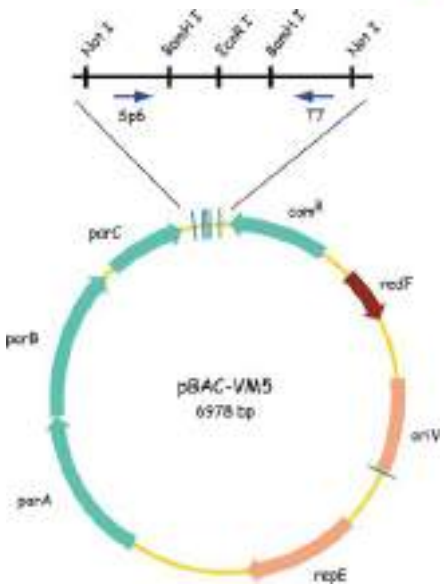
Key features:

- Allow **high-level protein synthesis** in bacteria such as *E. coli*.
- Sometimes used in *Bacillus subtilis*, which can better secrete proteins.
- Can also be delivered into **mammalian cells** using viral systems like **SV40**.
- Useful in **human gene therapy** applications.

However, not all eukaryotic proteins fold or function correctly in bacteria because bacterial cells lack the organelles needed for protein modification.

4. **Bacterial Artificial Chromosomes (BACs)**

BACs are large, low-copy-number plasmids derived from bacterial **F-factors**, which control replication. They are ideal for cloning very large DNA fragments, especially in genome projects.

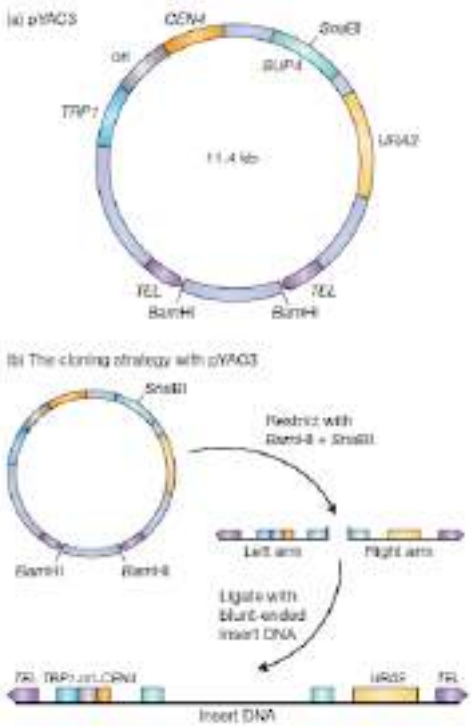


Key features:

- Can accept inserts up to **100–300 kb** in size.
- Maintain only one or two copies per cell for stability.
- Played an important role in the **Human Genome Project**.

5. **Yeast Artificial Chromosomes (YACs)**

YACs act as miniature versions of eukaryotic chromosomes and are used to clone very large DNA fragments. They contain all essential elements of a real chromosome, including **telomeres**, a **centromere**, and an **origin of replication**.



Key features:

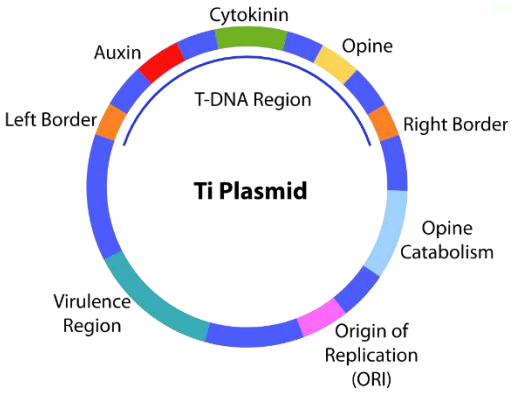
- Can carry DNA fragments from **200 kb up to 2 megabases** (**Mb**).
- Function like normal chromosomes inside **yeast cells**.
- Helped scientists clone and map human chromosomes during the **Human Genome Project**.

6. **Ti Vectors (Tumor-Inducing Plasmids)**

Ti vectors come from the soil bacterium *Rhizobium radiobacter* (formerly *Agrobacterium tumefaciens*), which naturally transfers DNA into plants. In nature, this bacterium causes **crown gall disease** by inserting a segment called **T-DNA** into a plant's genome, leading to tumor-like growths.

Key features:

- Modified to remove harmful genes like **auxin synthesis** genes.
- Used to **insert beneficial genes into plants**.
- Essential in **plant genetic engineering** and the development of **genetically modified crops**.



Each vector type has unique properties that make it suitable for specific purposes. For instance:

- Plasmids are best for **small DNA inserts**.
- Phage and cosmid vectors handle **medium-sized DNA fragments**.
- BACs and YACs carry **large genomic fragments**.
- Ti vectors are used for **plant transformation**.
- Expression vectors specialize in **protein production**.

Together, these vector systems have allowed scientists to manipulate genes in bacteria, plants, and even humans, advancing our understanding of genetics and enabling breakthroughs in medicine, agriculture, and biotechnology.

TABLE 2 A COMPARISON OF DNA VECTORS AND THEIR APPLICATIONS

Vector Type	Maximum Insert Size (kb)	Applications	Limitations
Bacterial plasmid vectors (circular)	~ 6–12	DNA cloning, protein expression, subcloning, direct sequencing of insert DNA	Restricted insert size; limited expression of proteins; copy number problems; replication restricted to bacteria
Bacteriophage vectors (linear)	~ 25	cDNA, genomic and expression libraries	Packaging limits DNA insert size; host replication problems
Cosmid (circular)	~ 35	cDNA and genomic libraries, cloning large DNA fragments	Phage packaging restrictions; not ideal for protein expression; cannot be replicated in mammalian cells
Bacterial artificial chromosome (BAC, circular)	~ 300	Genomic libraries, cloning large DNA fragments	Replication restricted to bacteria; cannot be used for protein expression
Yeast artificial chromosome (YAC, circular)	200–2,000	Genomic libraries, cloning large DNA fragments	Must be grown in yeast; cannot be used in bacteria
Ti vector (circular)	Varies depending on type of Ti vector used	Gene transfer in plants	Limited to use in plant cells only; number of restriction sites randomly distributed; large size of vector not easily manipulated

HOW DO YOU IDENTIFY AND CLONE A GENE OF INTEREST?

In molecular biology, **cutting and pasting DNA pieces** to create recombinant molecules has become a common practice. However, earlier cloning methods often relied on **random cloning**—scientists would cut large DNA samples using restriction enzymes and insert fragments into plasmids without knowing which specific gene was being cloned. This process, known as "shotgun cloning," produces thousands of random DNA combinations. If scientists wanted to isolate a particular gene, such as the insulin gene, it would be extremely difficult to identify which recombinant plasmid actually contained it.

To solve this problem, scientists developed a more organized approach using DNA libraries. DNA libraries store large collections of DNA fragments or genes from an organism, making it possible to search for and isolate a specific gene of interest.

Creating DNA Libraries

A **DNA library** is a collection of cloned DNA fragments from an organism, stored in bacteria or viruses as hosts. These libraries can be preserved for long periods and used repeatedly to identify genes. There are two main types of DNA libraries:

1. **Genomic DNA Libraries**

A **genomic library** contains the entire DNA sequence of an organism, including both coding and non-coding regions.

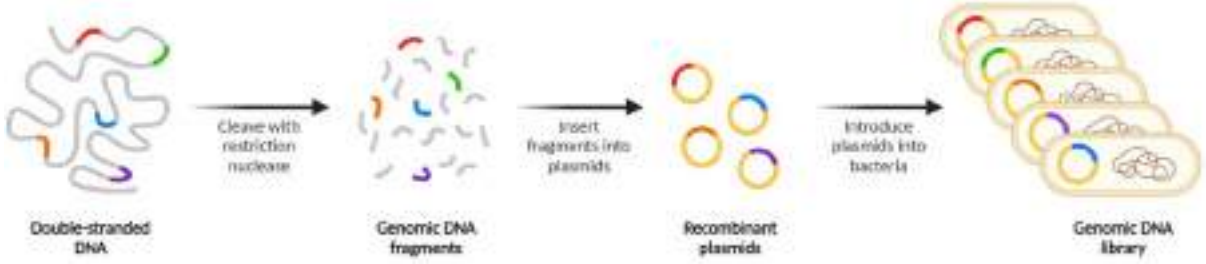
How it is made:

- DNA is extracted from the organism's cells.
- The DNA is **cut into fragments** using restriction enzymes.
- These fragments are **inserted into vectors** (such as plasmids, BACs, or YACs) using **DNA ligase**.
- The recombinant vectors are introduced into **bacteria**, and each bacterial colony contains a different DNA fragment.

Each bacterial colony represents one "book" in the library, and together they form the complete "genome collection."

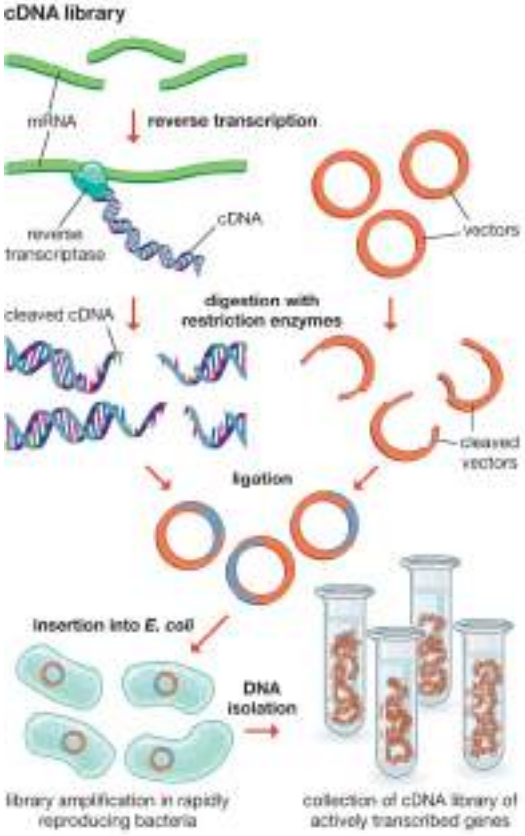
Key characteristics:

- Contains both **exons** (coding DNA) and **introns** (non-coding DNA).
- Useful for studying the structure and organization of entire genomes.
- Searching for a single gene is challenging because of the huge amount of DNA present, making it like "looking for a needle in a haystack."



2. Complementary DNA (cDNA) Libraries

A **cDNA library** represents only the **expressed genes** (the ones being actively used by the cell). Instead of starting with chromosomal DNA, scientists use **messenger RNA** (**mRNA**) from the cells or tissue of interest.



How it is made:

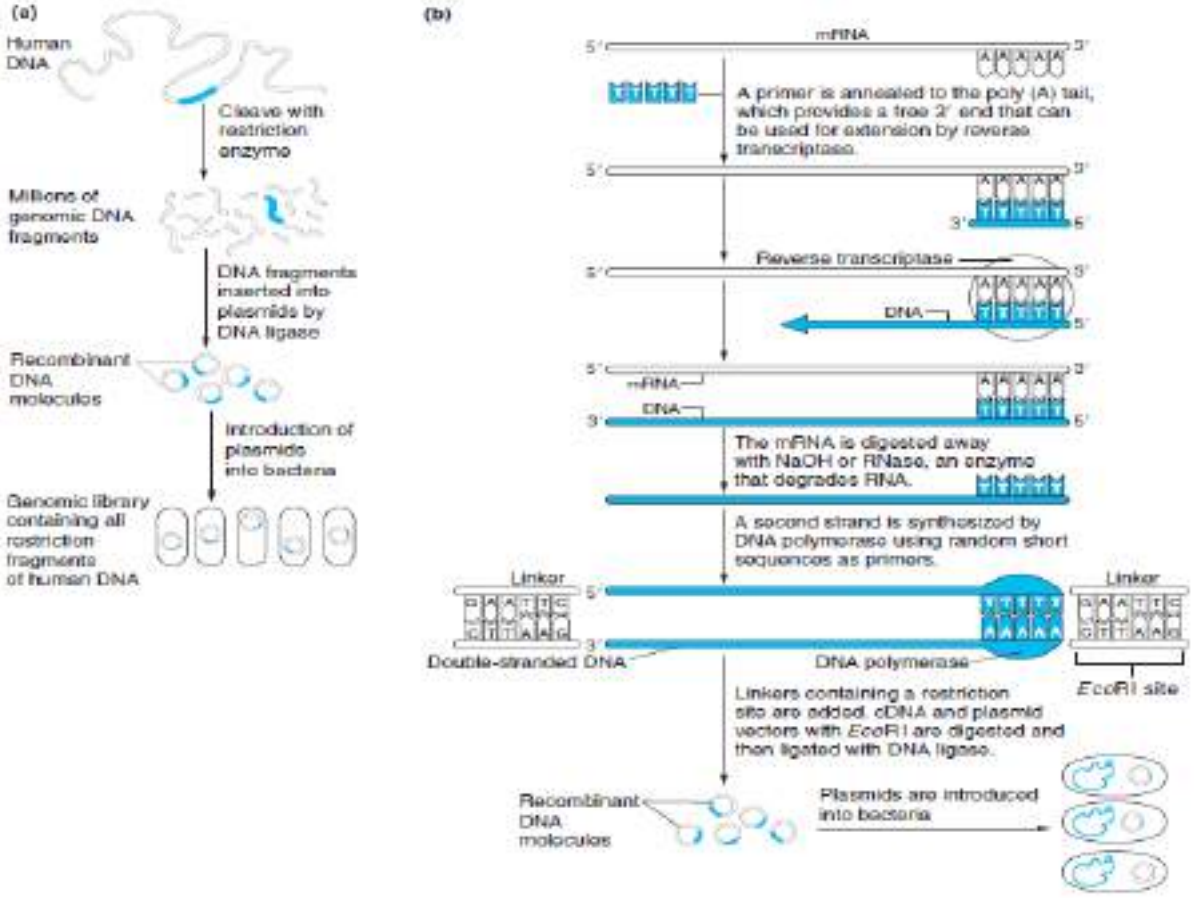
- mRNA is extracted from cells.
- The enzyme **reverse transcriptase** (RT) converts mRNA into single-stranded DNA.
- A second strand is synthesized using **DNA polymerase**, creating double-stranded complementary DNA (cDNA).
- Short DNA sequences called **linkers** (which contain restriction sites) are added to the ends of the cDNA.
- These cDNA molecules are inserted into plasmids and introduced into bacteria.

Key characteristics:

- Represents **only the genes that are active** in a particular tissue or condition.
- **Introns are not included**, so the sequences are simpler and easier to express in bacteria.
- Ideal for producing **eukaryotic proteins** and studying **gene expression**.
- Libraries can be made from specific tissues, such as hormone-stimulated cells, to target genes that are only expressed under certain conditions.

Advantages of cDNA Libraries:

- Contain only **expressed genes**, making them efficient for studying protein-coding regions.
- Useful when researchers want to produce eukaryotic proteins in bacteria.
- Easier to handle compared to genomic libraries because they exclude introns.



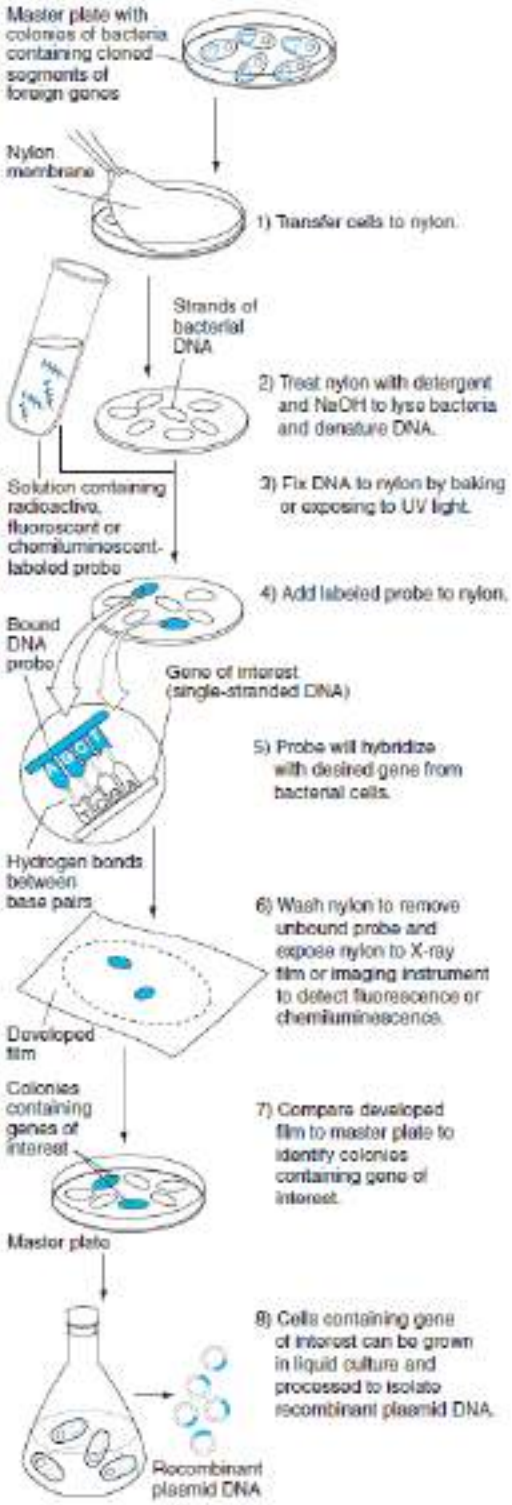
Library Screening: Finding the Gene of Interest

Once a DNA library has been created, scientists must identify which bacterial colony or viral clone contains the gene they want. This process is called **library screening**.

One common technique is **colony hybridization**, which works as follows:

Steps in colony hybridization:

1. Bacterial colonies containing recombinant DNA are grown on agar plates.
2. A **nylon** or **nitrocellulose membrane** is placed over the colonies to pick up a sample of each one.
3. The membrane is treated with an **alkaline solution** to break open the cells and separate DNA strands.
4. A **DNA probe** — a short, single-stranded DNA sequence that is complementary to the gene of interest — is added.
5. The probe binds (hybridizes) to its matching sequence on the membrane.
6. The probe is labeled with a **radioactive tag** or **fluorescent dye**, allowing scientists to detect where it has bound.
7. The membrane is washed, and the labeled spots show which colonies contain the target gene.



This process is called **autoradiography** (if radioactive probes are used) or fluorescent imaging (if fluorescent probes are used). The glowing or dark spots indicate colonies that contain the desired recombinant DNA.

From Libraries to Genomics

With the rise of **whole-genome sequencing technologies**, scientists can now identify all genes within a genome directly, without creating libraries one by one. This approach, known as **whole-genome shotgun sequencing**, allows researchers to study genes much faster and with greater accuracy.

However, understanding DNA libraries remains important because they serve as the foundation for many genetic discoveries and still provide valuable resources for cloning, sequencing, and gene expression studies.